**Q-Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers.

Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.**

To use the **Simulation Mode** in Cisco Packet Tracer to capture and analyze **RIP (Routing Information Protocol) update packets** exchanged between two routers and identify the source, destination, advertised routes, and routing metrics.

To configure a network with two routers.
To understand the operation of RIP.
To observe RIP update packets in Cisco Packet Tracer Simulation Mode.
To identify the source and destination of RIP packets.
To examine the routes advertised by each router.
To understand how routers exchange routing information dynamically.

Topology

```
              Serial Link
       Se0/3/0              Se0/3/0
   +-----------+            +-----------+
   | Router0   |------------| Router1   |
   +-----------+            +-----------+
       G0/0                     G0/1
        |                        |
      Switch0                  Switch1
        |                        |
       PC0                      PC1
```

Configuration

RIP was configured on both routers so that they could exchange routing information.

Router0

Router0(config)# router rip
Router0(config-router)# version 2
Router0(config-router)# network <Router0-LAN-Network>
Router0(config-router)# network <Serial-Network>
Router0(config-router)# no auto-summary

Router1

Router1(config)# router rip
Router1(config-router)# version 2
Router1(config-router)# network <Router1-LAN-Network>

Router1(config-router)# network <Serial-Network>
Router1(config-router)# no auto-summary

The actual network addresses depend on the IP addressing used in the Packet Tracer topology.

7. Procedure

1. Created the network topology using two routers, two switches, and two PCs.
2. Connected Router0 and Router1 using their serial interfaces.
3. Connected each router to a switch.
4. Connected each switch to a PC.
5. Configured IP addresses on the router interfaces and PCs.
6. Configured RIP version 2 on both routers.
7. Verified that the router interfaces were operational.
8. Opened **Simulation Mode** in Cisco Packet Tracer.
9. Selected **Edit Filters**.
10. Removed unnecessary protocols and selected **RIP**.
11. Used **Capture/Forward** to observe the packets exchanged between Router0 and Router1.
12. Opened the RIP packet and examined the PDU details.
13. Recorded the source, destination, advertised networks, and metric.

8. RIP Packet Analysis

During the simulation, RIP update packets were observed traveling between the two routers.

RIP Update from Router0 to Router1

Parameter	Observation
Protocol	RIP
Source	Router0
Destination	Router1 / RIP destination
Advertised Route	<Router0-LAN-Network>
Metric	<Metric shown in PDU>

Router0 advertised the network connected to its LAN so that Router1 could learn how to reach that network.

RIP Update from Router1 to Router0

Parameter	Observation
Protocol	RIP

Source	Router1
Destination	Router0 / RIP destination
Advertised Route	<Router1-LAN-Network>
Metric	<Metric shown in PDU>

Router1 advertised its LAN network so that Router0 could learn how to reach the network connected to Router1.

Simulation Observation

The RIP packets were successfully observed in **Simulation Mode**.

The general packet exchange was:

Router0 ——— RIP Update ———▶ Router1

Router1 ——— RIP Update ———▶ Router0

The routers exchanged information about the networks that they knew about. The advertised routes were used by the receiving router to update its routing table.

RIP is a **distance-vector routing protocol**.

RIP uses **hop count** as its routing metric.

- A directly connected network has a metric of **0** from the router's perspective.
- A route learned through one neighboring router normally has a hop count of **1**.
- The maximum usable RIP hop count is **15**.
- A hop count of **16 represents an unreachable network**.

RIP version 2 also supports classless routing information and uses multicast address **224.0.0.9** for RIP updates.

RIP update packets were successfully captured and analyzed in Cisco Packet Tracer Simulation Mode. The packets were exchanged between Router0 and Router1 and contained information about reachable networks. Router0 advertised its connected LAN network to Router1, while Router1 advertised its connected LAN network to Router0. RIP uses hop count as its routing metric, and the receiving routers use the advertised information to update their routing tables. Therefore, the experiment successfully demonstrated dynamic route exchange using RIP.